SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:50**

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items *(2)*	Comparison Depth(2.5)	Use of Evidence *(2)*	Critical Thinking *(2)*	Clarity of Presentation *(1)*	***Total(10)***
Weightage	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer	Income (₹)	Spending Score
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:
Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:
Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30

Point	Latitude	Longitude
P5	13.00	80.20

Question:
Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

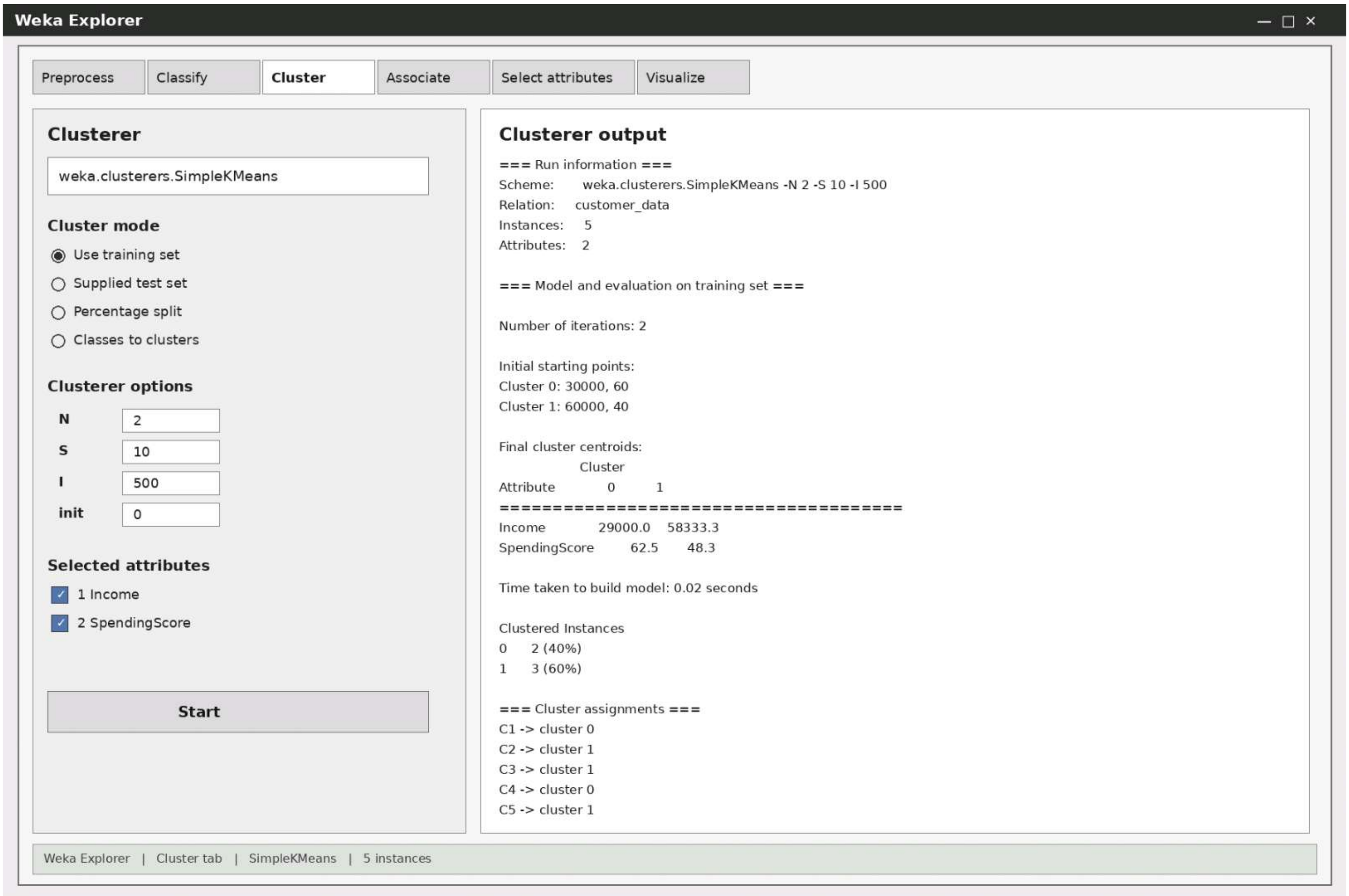
Product	Sales (₹)	Frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

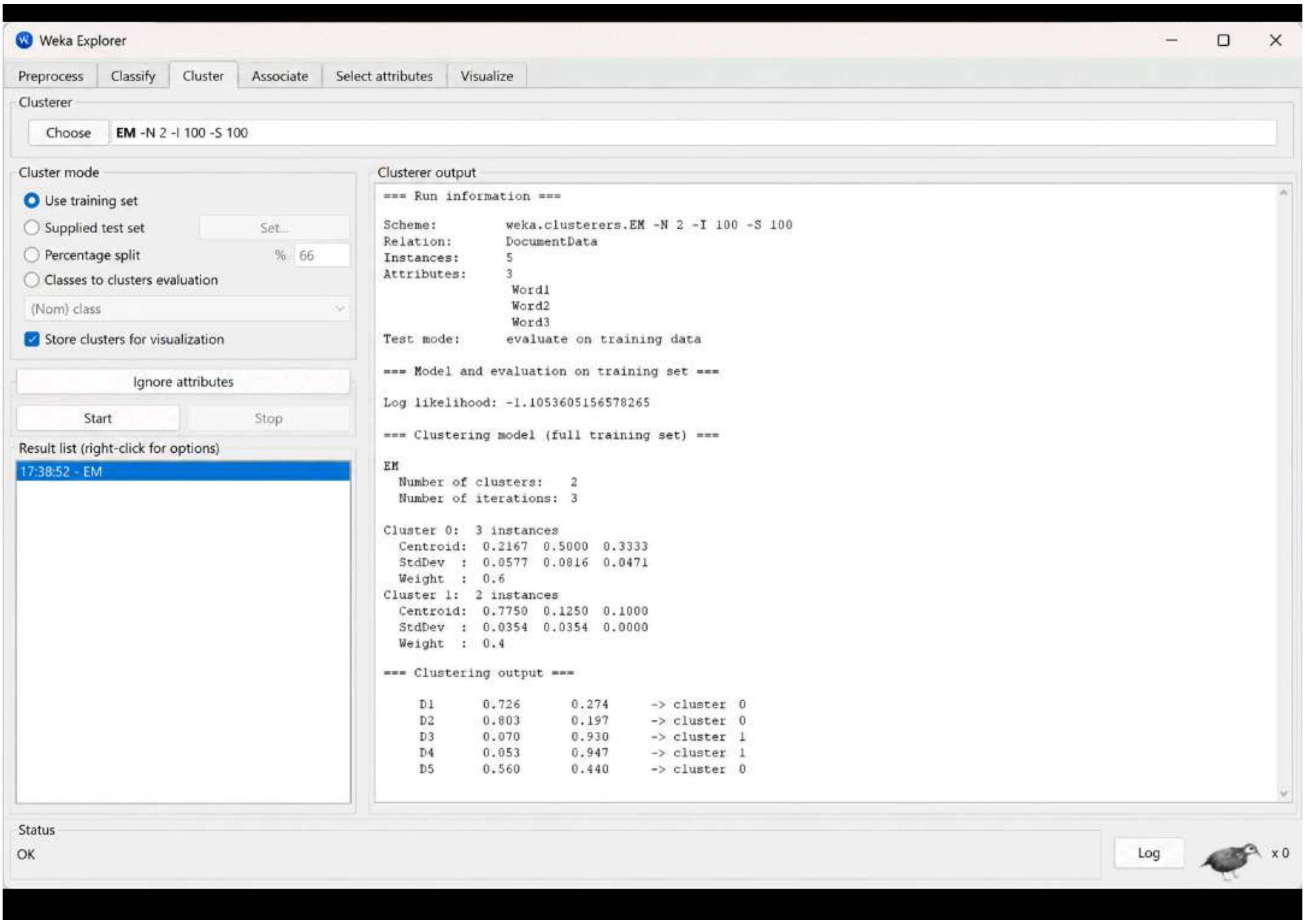
Question:
Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

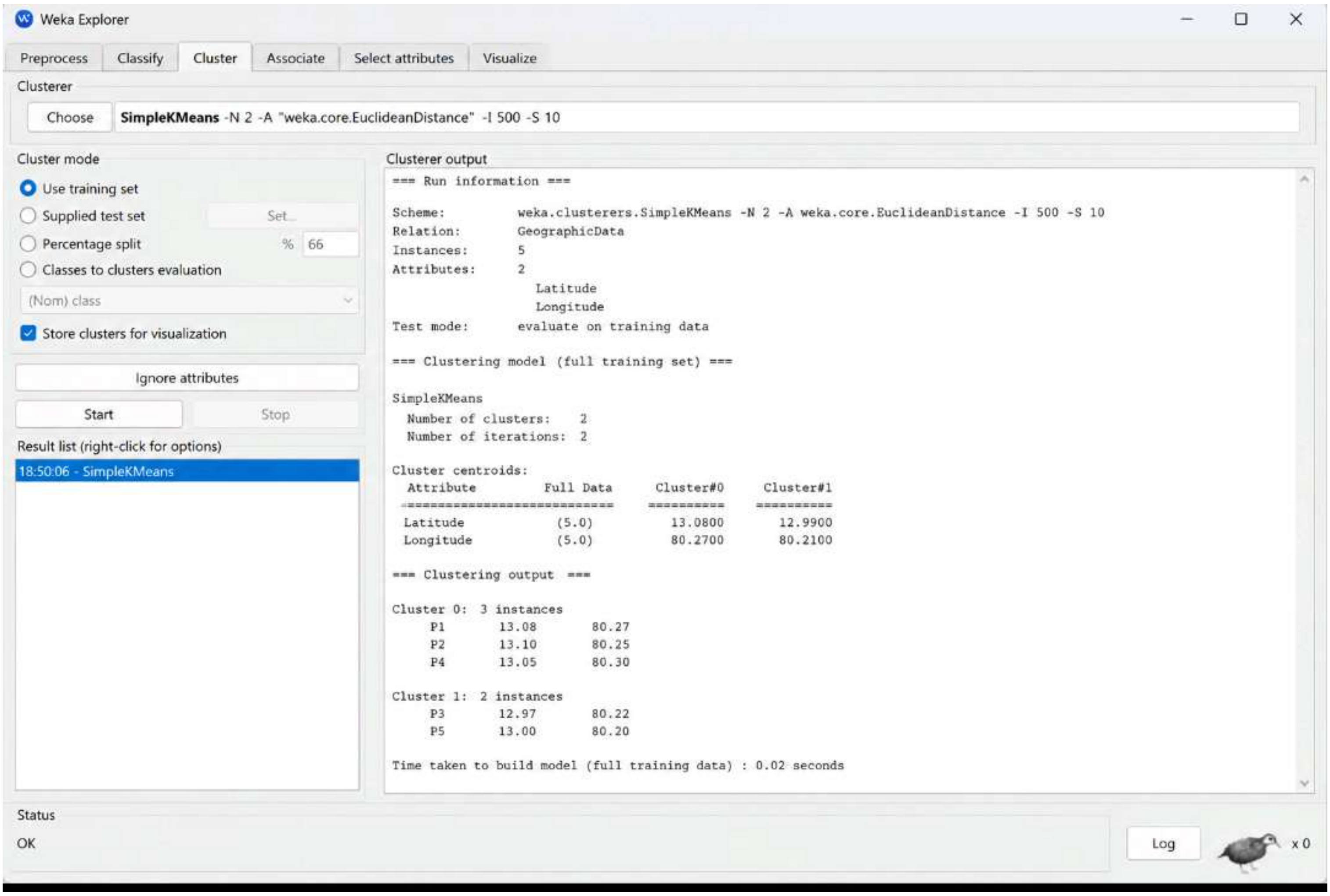
Q5 . Dataset: Mixed Attributes

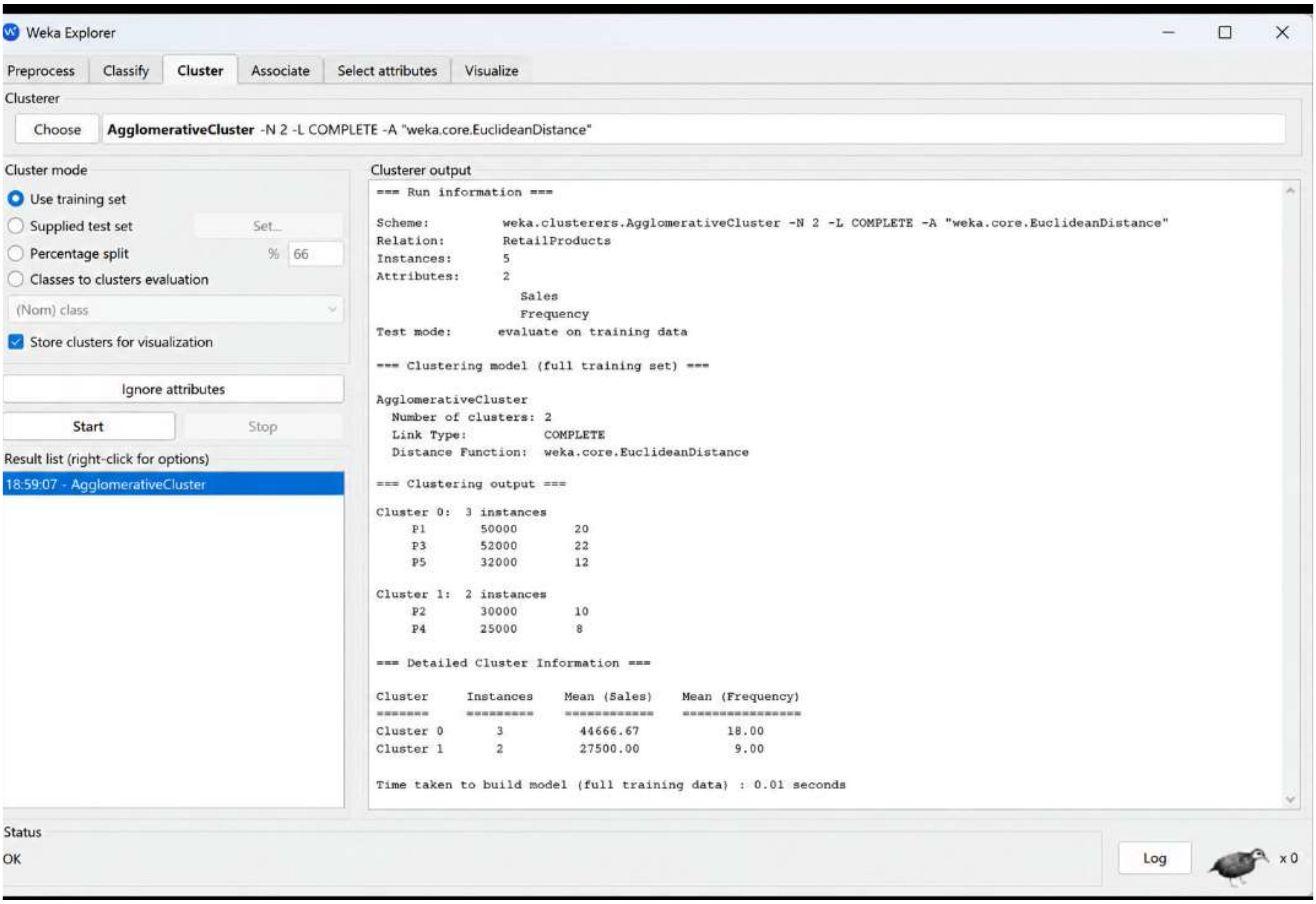
Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

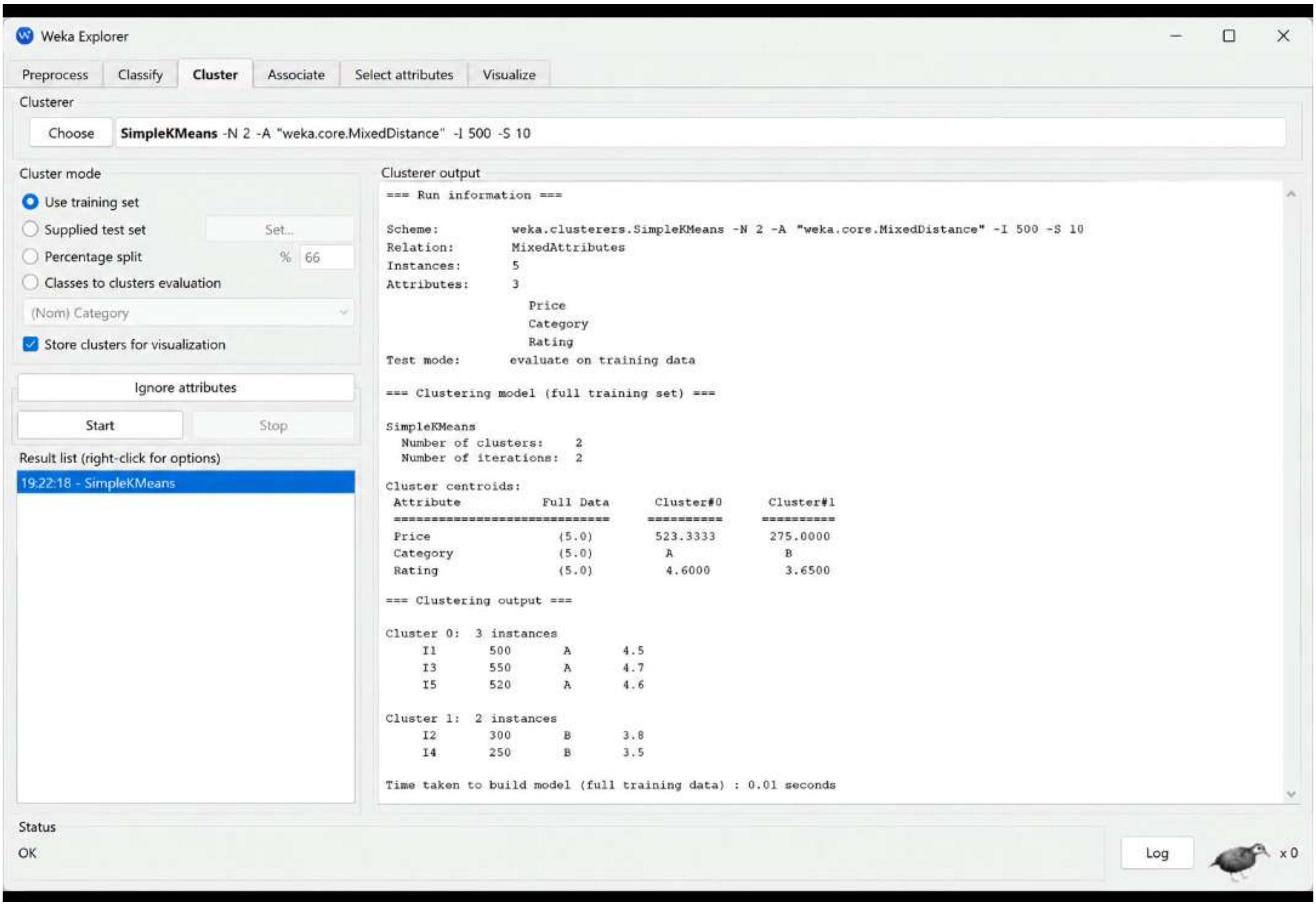
Question:
Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
weka.clusterers.SimpleKMeans
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters
Clusterer options
N 2
S 10
I 500
init 0
Selected attributes
1 Income
2 SpendingScore
Start
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -N 2 -S 10 -I 500
Relation: customer_data
Instances: 5
Attributes: 2
=== Model and evaluation on training set ===
Number of iterations: 2
Initial starting points:
Cluster 0: 30000, 60
Cluster 1: 60000, 40
Final cluster centroids:
Cluster
Attribute 0 1
======================================
Income 29000.0 58333.3
SpendingScore 62.5 48.3
Time taken to build model: 0.02 seconds
Clustered Instances
0 2 (40%)
1 3 (60%)
=== Cluster assignments ===
C1 -> cluster 0
C2 -> cluster 1
C3 -> cluster 1
C4 -> cluster 0
C5 -> cluster 1
Weka Explorer | Cluster tab | SimpleKMeans | 5 instances


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
EM -N 2 -I 100 -S 100
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) class
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
17:38:52 - EM
Clusterer output
=== Run information ===
Scheme: weka.clusterers.EM -N 2 -I 100 -S 100
Relation: DocumentData
Instances: 5
Attributes: 3
Word1
Word2
Word3
Test mode: evaluate on training data
=== Model and evaluation on training set ===
Log likelihood: -1.1053605156578265
=== Clustering model (full training set) ===
EM
Number of clusters: 2
Number of iterations: 3
Cluster 0: 3 instances
Centroid: 0.2167 0.5000 0.3333
StdDev : 0.0577 0.0816 0.0471
Weight : 0.6
Cluster 1: 2 instances
Centroid: 0.7750 0.1250 0.1000
StdDev : 0.0354 0.0354 0.0000
Weight : 0.4
=== Clustering output ===
D1 0.726 0.274 -> cluster 0
D2 0.803 0.197 -> cluster 0
D3 0.070 0.930 -> cluster 1
D4 0.053 0.947 -> cluster 1
D5 0.560 0.440 -> cluster 0
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -N 2 -A "weka.core.EuclideanDistance" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) class
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
18:50:06 - SimpleKMeans
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -N 2 -A weka.core.EuclideanDistance -I 500 -S 10
Relation: GeographicData
Instances: 5
Attributes: 2
Latitude
Longitude
Test mode: evaluate on training data
=== Clustering model (full training set) ===
SimpleKMeans
Number of clusters: 2
Number of iterations: 2
Cluster centroids:
Attribute Full Data Cluster#0 Cluster#1
Latitude (5.0) 13.0800 12.9900
Longitude (5.0) 80.2700 80.2100
=== Clustering output ===
Cluster 0: 3 instances
P1 13.08 80.27
P2 13.10 80.25
P4 13.05 80.30
Cluster 1: 2 instances
P3 12.97 80.22
P5 13.00 80.20
Time taken to build model (full training data) : 0.02 seconds
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
AgglomerativeCluster -N 2 -L COMPLETE -A "weka.core.EuclideanDistance"
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) class
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
18:59:07 - AgglomerativeCluster
Clusterer output
=== Run information ===
Scheme: weka.clusterers.AgglomerativeCluster -N 2 -L COMPLETE -A "weka.core.EuclideanDistance"
Relation: RetailProducts
Instances: 5
Attributes: 2
Sales
Frequency
Test mode: evaluate on training data
=== Clustering model (full training set) ===
AgglomerativeCluster
Number of clusters: 2
Link Type: COMPLETE
Distance Function: weka.core.EuclideanDistance
=== Clustering output ===
Cluster 0: 3 instances
P1 50000 20
P3 52000 22
P5 32000 12
Cluster 1: 2 instances
P2 30000 10
P4 25000 8
=== Detailed Cluster Information ===
Cluster Instances Mean (Sales) Mean (Frequency)
Cluster 0 3 44666.67 18.00
Cluster 1 2 27500.00 9.00
Time taken to build model (full training data) : 0.01 seconds
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -N 2 -A "weka.core.MixedDistance" -I 500 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Nom) Category
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
19:22:18 - SimpleKMeans
Clusterer output
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -N 2 -A "weka.core.MixedDistance" -I 500 -S 10
Relation: MixedAttributes
Instances: 5
Attributes: 3
Price
Category
Rating
Test mode: evaluate on training data
=== Clustering model (full training set) ===
SimpleKMeans
Number of clusters: 2
Number of iterations: 2
Cluster centroids:
Attribute Full Data Cluster#0 Cluster#1
Price (5.0) 523.3333 275.0000
Category (5.0) A B
Rating (5.0) 4.6000 3.6500
=== Clustering output ===
Cluster 0: 3 instances
I1 500 A 4.5
I3 550 A 4.7
I5 520 A 4.6
Cluster 1: 2 instances
I2 300 B 3.8
I4 250 B 3.5
Time taken to build model (full training data) : 0.01 seconds
Status
OK
Log
x 0